

ARUNKUMAR S

Chennai, Tamil Nadu, India

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Education

Bachelor of Technology in Smart Manufacturing

August 2023 – May 2027

Indian Institute of Information Technology Design & Manufacturing

7.4/10

HSC, Class XII

Sept 2021 – May 2023

Narayana Junior College, Nellore

96%

SSC, Class X

June 2020 – April 2021

Narayana Higher Secondary School, Nellore

99%

Technical Skills

Languages: Python, C, HTML, SQL

Design & CAD Tools: SolidWorks, AutoCAD, Fusion 360

Simulation & Analysis: ANSYS, LS-Dyna, Hypermesh

Engineering Skills: 3D Modelling, GD&T, 2D Documentation, FEA, Product Design, Prototyping

Experience

Eaura Pvt Ltd (Government Approved Startup)

June 2025 – Present

Founder / CEO

Chennai, Tamil Nadu / Robotics & Automation

- Founded a government-approved startup developing a personal voice assistant to help senior citizens manage daily activities and support independent living.
- Leading end-to-end MVP development including AI features for reminders, emergency support, smart scheduling, system architecture, and hardware/software integration.

SRJ Energy Technologies (IITM Research Park)

April 2025 – Present

Research Intern

- Working on high-capacity cylindrical battery cell and composite battery enclosure design for EV applications, targeting improved strength-to-weight ratio and impact resistance.
- Performed thermal and structural optimization using SolidWorks and ANSYS, while developing validation protocols to enhance safety, durability, and heat dissipation.

Neyveli Lignite Corporation India Limited

May 2025 – Present

Project Research Fellow

- Designed a lignite separation and conveyor handling system, reducing operational downtime by **20%** through improved workflow and optimized equipment layout.
- Integrated magnetic separation and vibrating screening mechanisms to improve stone segregation accuracy, reduce manual sorting, and enhance plant safety and efficiency.

Projects

Mobile Banana Processing Unit | *Trident Fruits, Mumbai*

Dec 2024 – Mar 2025

- Designed a mobile banana processing and handling unit in SolidWorks to support efficient post-harvest operations including de-handling, cleaning, and packaging.
- Developed a multi-stage quality control workflow incorporating fungicide washing, grading, labeling, and boxing to enhance shelf life and reduce spoilage during transport.
- Optimized the unit layout for portability, improved processing throughput, and suitability for on-site agricultural deployment.

Driving License Authentication & Driver Monitoring System | *Course work*

Jan 2025 – May 2025

- Developed an intelligent driver authorization system integrating fingerprint authentication and RFID-based license verification to ensure only eligible, licensed, and verified individuals can access, start, and operate the vehicle securely.
- Implemented continuous facial recognition monitoring during vehicle operation to verify driver identity in real time, detect unauthorized access attempts, and enhance overall operational safety.

- Leading the development of a farm-based transport E-vehicle designed to carry banana bunches from inside the plantation to the loading area.
- Head of the Vehicle Dynamics subsystem, responsible for suspension design, steering behaviour, braking performance, stability, and ride analysis on uneven terrain.
- Coordinating with powertrain, chassis, and electronics teams to ensure integrated performance, maneuverability, and load-carrying efficiency.

Research Publications

Data-Driven Mechanical Characterization of Re-Entrant Auxetic Honeycomb Structures Using Polynomial Ridge Regression

ASME IMECE-INDIA 2026

Accepted

- Developed an interpretable Polynomial Ridge Regression (PolyRidge) model to predict specific plateau stress (SPS), Poisson’s ratio, and specific energy absorption (SEA) of re-entrant auxetic honeycomb structures using geometric parameters (θ , H/L).
- Achieved high predictive accuracy with R^2 values of 0.9455–0.9962 and integrated the model into a GUI for rapid design-oriented performance prediction, reducing reliance on repeated experiments and FEA simulations.

AI-Assisted Rock Detection & Mechanical Diversion System for Lignite Mining Conveyor Operations

ICoRD 27 – International Conference on Research into Design

Under Review

- Designed a real-time AI-assisted rock separation system using a YOLOv11-based vision model with a synchronized pneumatic diversion mechanism for lignite conveyor operations.
- Developed a multi-stage screening and AI detection pipeline to divert oversized rock fragments without stopping belt operation, reducing crusher downtime and worker exposure to hazardous environments.

Low Velocity Impact Response of Advanced Composites

Journal Review Paper

Under Review

- Authored a comprehensive review on low-velocity impact damage mechanisms in advanced composites, focusing on barely visible impact damage (BVID), matrix cracking, delamination, and energy absorption in natural, synthetic, and hybrid laminates.
- Analyzed state-of-the-art mitigation and modelling approaches including nanomodified composites, bio-inspired architectures, self-healing systems, and FEA-based delamination prediction using Cohesive Zone Models (CZM) and failure criteria (Hashin/Puck).

Position of Responsibility

Department Lead, Placement Cell

Dec 2024 – Present

- Coordinated campus recruitment drives by managing communication with recruiters, faculty, and students, while organizing pre-placement talks, interviews, and assessments.
- Maintained student placement records and led volunteer teams to support smooth execution of recruitment events.

Team Lead, Smart India Hackathon

Nov 2025

- Led a 6-member team in proposing a Solar-powered Dewatering system for mining operations, integrating PV panels, battery storage, and inverter-controlled pump control for sustainable performance.
- Directed the development of an automation-focused concept using IoT sensors and SCADA-based monitoring to ensure uninterrupted dewatering, reduce diesel/electricity costs, and enhance safety in harsh mining environments.

Achievements & Certifications

- **GATE 2026 – Production & Industrial Engineering (PI)** 2026
Score: **384** | All India Rank (AIR): **710**

Extra-Curricular Activities

- **Department Volleyball Team Captain**, Department of Mechanical Engineering 2025
- **Captain**, School Athletics & Sports Team 2023
- **Frystgen Tech Hackathon**, VIT Amravati Oct 2025